Positional Encodings Anchor Spatial Structure in Vision Transformers: A Geometric Perspective on Robustness

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Abstract

Positional embeddings (PEs) in Vision Transformers (ViTs) are known to impact performance and robustness, but their role in shaping internal spatial representations is not well understood. In this work, we study how different forms of PEs influence the representational geometry of ViTs and its relationship to robustness. We introduce a metric, the Spatial Similarity Distance Correlation (SSDC), to quantify spatial structure in token representations. Using this metric, we show that ViTs trained without PEs still develop non-trivial spatial structure, but this structure is driven by visual content and collapses under token permutation. In contrast, we find that all PEs considered (learned absolute, sinusoidal, and rotary) induce a consistent shift toward an index-anchored spatial organization. Representations in these models remain stable under perturbations that disrupt content, and exhibit substantially improved robustness to distributional shifts. We further show that while different PEs produce distinct depth-wise trajectories of spatial structure, their robustness properties are largely similar, suggesting that robustness depends on the presence of a stable positional reference frame more than it depends on the specific encoding mechanism. These results offer a mechanistic account of how positional encodings shape internal representations, with implications for the principled design of future encoding schemes.

1 Introduction

Vision Transformers (ViTs) model images as sequences of patch tokens processed by self-attention [Dosovitskiy et al., 2021]. Unlike convolutional architectures, they lack built-in inductive biases toward locality and translation equivariance, and instead rely on positional embeddings (PEs) to inject spatial information, enabling the model to distinguish tokens originating from different locations.

While PEs are designed to provide positional information, this design does not determine how that signal is integrated into internal representations. In particular, it remains unclear whether positional information organizes token representations into similarity structures anchored to absolute indices, or whether spatial structure continues to arise primarily from visual content.

Prior work shows that ViTs retain substantial performance even when positional information is removed or degraded [Dosovitskiy et al., 2021, Chu et al., 2023], suggesting that spatial relationships can partially emerge from patch content alone. This raises a central question: if spatial structure can arise without explicit positional guidance, what functional role do positional embeddings play?

Existing studies have largely addressed this question through downstream performance comparisons or architectural variations. While informative, these approaches provide limited insight into how positional information shapes internal representations. In particular, it remains unclear whether

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different positional encoding schemes (learned absolute, sinusoidal, or rotary) induce distinct spatial reasoning strategies, or whether their effects on robustness arise from a shared mechanism.

In this work, we adopt a mechanistic perspective. We analyze the evolution of token representations across the transformer stack using tools from representational geometry [Raghu et al., 2021], introducing the Spatial Similarity Distance Correlation (SSDC) as a probe of spatial structure. Critically, we use SSDC in conjunction with a random permutation intervention at inference to distinguish whether spatial organization is anchored to token indices or driven by patch content. We compare models trained with learned absolute positional embeddings (APE), sinusoidal encodings, rotary embeddings (RoPE), and no positional embeddings, and evaluate their robustness to distributional shifts.

Our central finding is that the specific encoding mechanism matters less than the presence of a consistent positional signal. We show that:

- **Positional encodings induce index-based spatial organization:** All PE types shift ViTs away from content-driven spatial structure toward representations anchored to token indices, as revealed by SSDC under token permutation. Models without a stable positional signal rely on patch content and exhibit SSDC collapse under permutation.
- **This shift, not the encoding form, confers robustness:** Despite differing in how spatial structure develops across depth, APE, sinusoidal, and RoPE models exhibit broadly comparable robustness to distributional shifts (despite consistent but smaller differences between encoding schemes), while models lacking index-based organization are substantially more fragile.
- **A stable positional reference frame is necessary:** Using Random Permutation Training (RPT), which preserves PEs but destroys index-to-location consistency, we show that robustness depends on a stable positional reference frame rather than the mere presence of positional embeddings.

Together, these results provide a unified, mechanistic account of how positional encodings shape internal representations and why they remain critical for robust visual recognition.

2 Related Work

Positional Information in Vision Transformers

The standard Vision Transformer (ViT) breaks the permutation invariance of self-attention by adding learnable absolute positional embeddings (PEs) to patch tokens [Dosovitskiy et al., 2021], establishing the dominant paradigm for spatial encoding. However, ViTs retain substantial performance when positional information is degraded or removed [Dosovitskiy et al., 2021, Chu et al., 2023], suggesting that spatial structure can partially emerge from patch content alone.

Similar observations have been reported beyond vision. Recent work on decoder-only transformers shows that models trained without PEs can recover positional information implicitly and tend to rely on relative positions in practice [Kazemnejad et al., 2023]. Earlier findings in convolutional networks further demonstrate that substantial positional information can be learned implicitly from architectural biases such as zero-padding [Islam* et al., 2020]. Together, these results suggest that explicit positional signals are not strictly required for structured spatial information to emerge.

This creates a central puzzle: if spatial structure can arise without explicit positional guidance, what functional role do PEs play? Prior work has primarily addressed this question through architectural variants [d'Ascoli et al., 2022, Liu et al., 2021, Heo et al., 2024] or performance comparisons [Dosovitskiy et al., 2021, Chu et al., 2023], leaving their mechanistic impact on internal representations largely unexplored.

Representational Analysis of Transformers

A separate line of work studies the geometry and dynamics of transformer representations. Early analyses compare ViT and CNN representations [Raghu et al., 2021], revealing differences in spatial organization. Subsequent work examines how attention transforms representations [Kobayashi et al., 2021], how representational rank evolves with depth [Dong et al., 2021], and how token representations tend to homogenize in deeper layers [Bhojanapalli et al., 2021]. The residual stream framework provides a useful lens for analyzing these dynamics [Elhage et al., 2021]. However,

these approaches do not isolate the causal role of positional embeddings, nor do they connect representational structure to robustness.

Robustness of Visual Models

Vision Transformers exhibit distinct robustness profiles compared to convolutional networks. Prior work shows that transformers are generally more robust to certain spatial perturbations but can be more sensitive to texture-based changes [Bhojanapalli et al., 2021]. Additional studies report favorable out-of-distribution generalization properties for ViTs [Paul and Chen, 2022], connecting to broader findings on shape versus texture bias in visual recognition [Geirhos et al., 2019]. While the impact of positional embeddings on robustness has been observed (particularly that models trained with PEs exhibit better robustness profiles than models trained without them) [Mao et al., 2021], the relationship between a model's spatial organization strategy (whether anchored to absolute position or inferred from content) and its robustness to distributional shifts remains poorly understood.

Our Contribution

We connect these lines of work by showing that positional embeddings induce a shift toward index-based spatial organization, and that this shift (rather than the specific encoding mechanism) is the dominant factor underlying robustness. Using SSDC and controlled permutation interventions (RPT and RPI), we provide a mechanistic account of how positional information shapes internal representations and why it improves robustness beyond merely breaking permutation invariance.

3 Preliminaries

3.1 Vision Transformer Architecture and Positional Encodings

All models are Vision Transformers trained from scratch on ImageNet-100, with approximately 22M parameters (details in Appendix A). Images are partitioned into fixed-size patches, projected into token embeddings, and processed by a stack of self-attention and feedforward layers.

Since self-attention is permutation invariant, positional encodings are required to inject spatial information. We consider three commonly used PE schemes, all adapted to 2D grids:

Learned Absolute Positional Embeddings (APE): learnable vectors added to token embeddings before the first transformer block, establishing a fixed index-to-location mapping.

Sinusoidal Positional Embeddings: fixed, deterministic encodings constructed from sinusoidal functions applied independently along spatial axes and added to token embeddings.

Rotary Positional Embeddings (RoPE): position-dependent rotations applied to query and key vectors within each attention layer, introducing positional information multiplicatively.

These approaches differ in parameterization (learned vs. fixed) and integration (additive vs. multiplicative), enabling comparison of how different positional signals shape internal representations.

3.2 Index-Based and Content-Based Spatial Organization

We distinguish between two qualitatvely distinct modes of spatial organization in token representations.

Index-based spatial organization refers to representations whose similarity structure depends systematically on token position. Tokens that are spatially proximate tend to have more similar representations by virtue of their indices, and this structure persists under disruptions to patch content. This definition is behavioral and does not assume explicit coordinate representations.

Content-based spatial organization refers to representations in which similarity is driven primarily by patch content. Spatial structure arises indirectly from natural image statistics and degrades under transformations that disrupt content or token ordering.

In practice, models may exhibit both behaviors; the key distinction is which signal dominates and whether spatial structure persists when content-based cues are disrupted.

4 Methods

4.1 Residual Stream Geometry

At selected layers, we extract the residual stream as a matrix $R \in \mathbb{R}^{T \times C}$, where T is the number of tokens and C the embedding dimension. We compute pairwise cosine similarities between unit-normalized token representations to form a symmetric similarity matrix, averaged across the batch dimension.

4.2 Spatial Similarity Distance Correlation

Let $S \in \mathbb{R}^{T \times T}$ denote the token similarity matrix, and let $p_i \in \mathbb{Z}^2$ denote the spatial coordinates of token i. Define the spatial distance matrix D by $D_{ij} = \|p_i - p_j\|_1$. We define SSDC as the Spearman rank correlation between similarity and negative spatial distance over all token pairs:

$$\mathrm{SSDC} = \rho_{\mathrm{Spearman}} \left(\{S_{ij}\}_{i<j},\ \{-D_{ij}\}_{i<j} \right).$$

Higher SSDC values indicate that spatially proximate tokens tend to have more similar representations. We use Spearman rank correlation to remain agnostic to the precise functional form relating spatial distance and representational similarity.

Critically, SSDC is not used as a standalone metric, absolute values are insufficient to characterize learned spatial organization (see Section 5.1). Instead, we use SSDC in two ways: to track its *evolution across depth* as a signature of learned spatial structure, and as a *permutation-sensitvity probe* via the Random Permutation at Inference (RPI) intervention described below.

4.3 Random Permutation at Inference (RPI)

To distinguish index-based from content-based organization, we randomly permute token order at inference while keeping positional indices fixed. This breaks the correspondence between token order and spatial location. Any SSDC recovery after permutation reflects reliance on absolute position rather than content.

4.4 Positional Embedding Magnitude Scaling

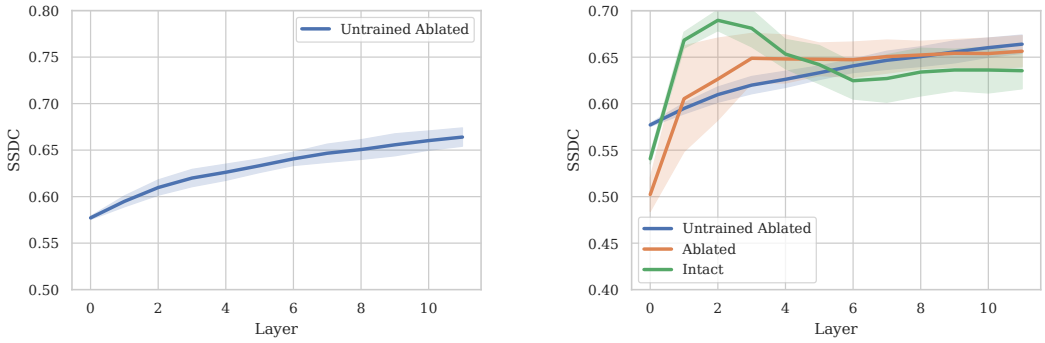
We scale positional embeddings at inference by a factor α, replacing $\mathbf{e}_i$ with $\alpha \mathbf{e}_i$. This provides a continuous intervention on positional signal strength without retraining. We apply this to APE and sinusoidal models; an equivalent scaling for RoPE is not directly defined due to its multiplicative formulation.

4.5 Fragility Score

We quantify robustness using the Fragility Score (FS):

$$\mathrm{FS} = 1 - \frac{A_{\mathrm{shift}}}{A_{\mathrm{normal}}},$$

where A_{normal} and A_{shift} denote accuracy on clean and shifted data. Higher values indicate greater sensitivity to distributional shift.



(a) **Evolution of SSDC across depth on untrained ablated models**

(b) **Evolution of SSDC across depth for on untrained ablated models, trained ablated models, and intact (trained with APE) models**

Figure 1: (a) SSDC grows weakly and remains at a relatively high value across layers, indicating static spatial correlations induced by architectural and data priors rather than learning.
(b) While untrained ablated models exhibit relatively high but slowly varying SSDC consistent with static data and architectural priors, trained ablated models display a sharp increase in early layers, indicating the emergence of learned spatial structure despite the absence of explicit positional encoding.

5 Results

5.1 Architectural Priors Induce Static Spatial Correlations at Initialization

Experimental Setup: We evaluate SSDC across all layers of untrained ablated models on the Imagenet-100 dataset. Unless stated otherwise, all reported results are averaged over 4 random seeds. Shaded regions in figures indicate variability across runs (±1 standard deviation).

Results: The untrained ablated model exhibits a substantial non-zero SSDC (approximately 0.57–0.64) with only a weak, gradual increase across depth (Figure 1a). This behavior is highly consistent across runs and reflects static spatial correlations induced by architectural priors and the inherent structure of natural images, rather than learned spatial reasoning.

Crucially, this baseline highlights that SSDC should not be interpreted as a standalone metric whose absolute magnitude reflects the presence or strength of learned spatial organization. Even in the absence of training, relatively high SSDC values emerge. Instead, the layer-wise dynamics of SSDC (in particular, the rate and pattern of change across depth) are the informative signal. In contrast to the shallow, nearly static progression observed here, trained models exhibit rapid and structured changes in SSDC (e.g., sharp increases in early layers), indicating the emergence of learned spatial structure.

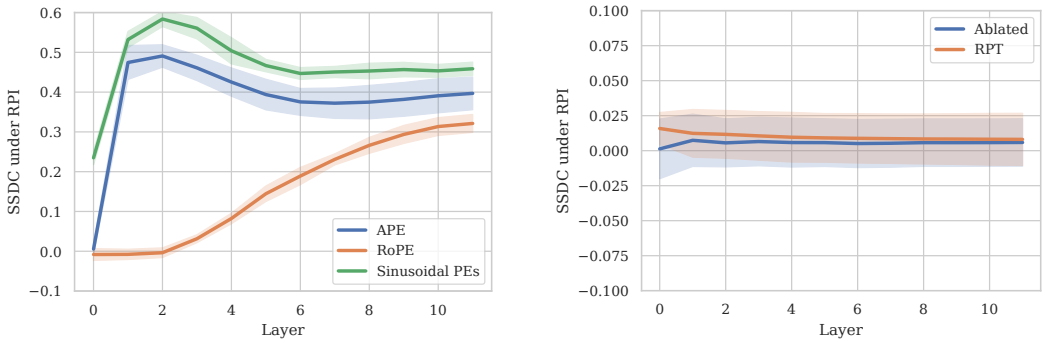
This establishes a static baseline, allowing us to distinguish genuinely learned spatial organization from correlations that arise purely from architectural and data-driven effects.

5.2 Emergence of Spatial Structure Without Positional Encoding

Experimental Setup: To investigate whether spatial structure can emerge in the absence of explicit positional information, we evaluate SSDC across all layers of untrained ablated models, trained ablated models, and trained intact (APE) models on the Imagenet-100 dataset.

Results: Figure 1b compares the layer-wise evolution of SSDC for three conditions: an untrained ablated model, a trained ablated model, and a trained model with positional embeddings. The untrained ablated model exhibits relatively high SSDC values (approximately 0.57–0.64) that increase only weakly with depth. As discussed in Section 5.1, this reflects static spatial correlations arising from architectural priors and the inherent structure of natural images, rather than learned spatial organization.

In contrast, the trained ablated model shows a markedly different behavior. Starting from a lower initial SSDC, it exhibits a rapid increase in early layers, followed by continued growth across depth.



(a) **Models with positional encodings.** APE, SinuPE, and RoPE models exhibit substantial SSDC recovery under RPI, indicating spatial organization anchored to token indices. RoPE shows a gradual accumulation of structure across depth, while APE and SinuPE exhibit earlier, injection-like peaks.

(b) **Models without a stable positional reference frame.** Ablated and RPT models collapse to near-zero SSDC across all layers under RPI, indicating that their spatial structure is entirely content-driven and does not survive token permutation.

Figure 2: **SSDC under random permutation at inference (RPI).** RPI disrupts the correspondence between token content and spatial position. Only models that anchor spatial structure to token indices exhibit SSDC recovery after permutation. In contrast, models lacking a consistent positional mapping collapse to near-zero SSDC, revealing a purely content-based spatial organization.

This dynamic pattern closely resembles the qualitative trajectory observed in the trained model with positional embeddings, despite the absence of any explicit positional signal. The key distinction is not the absolute magnitude of SSDC, but the rate and structure of its evolution: whereas the untrained model shows only a shallow, near-static progression, the trained ablated model displays sharp, structured changes indicative of learning.

These observations suggest that training induces non-trivial spatial structure even without positional embeddings. This interpretation is further supported by the non-trivial performance of ablated models and by prior findings that transformers can recover positional information implicitly. Taken together, the evidence indicates that spatial organization can emerge even without an explicit positional signal.

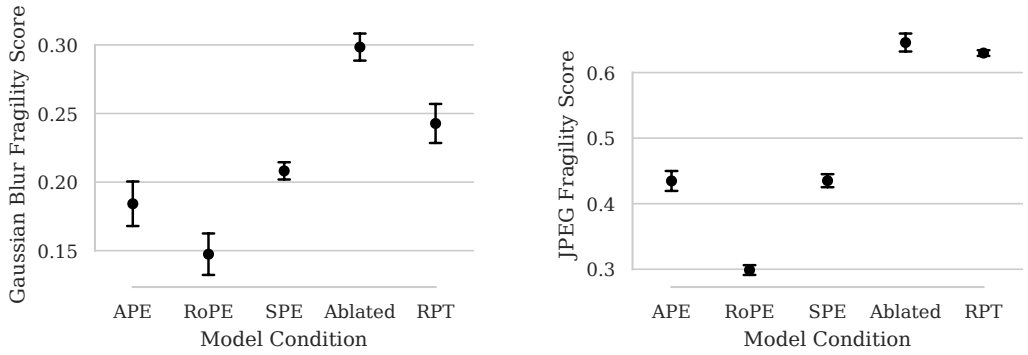
Importantly, we do not claim that this emergent structure is equivalent in nature or quality to that induced by positional embeddings. Rather, this result establishes that spatial structure can arise without explicit positional signals, motivating the need for a more precise characterization of its underlying mechanism, which we address in the following section.

5.3 Disentangling Index-Based and Content-Based Spatial Organization

Experimental Setup: To distinguish between index-based and content-based spatial organization, we evaluate SSDC across all layers under a *Random Permutation at Inference* (RPI) intervention. Concretely, patch tokens are randomly permuted before being processed by the transformer, while positional embedding indices (when present) remain fixed to their original spatial locations. This operation disrupts the correspondence between token content and spatial position, while preserving any mapping between token indices and positional embeddings.

Under this setup, any spatial structure that arises purely from patch content is destroyed, as spatially adjacent tokens no longer correspond to neighboring image patches. In contrast, if a model has learned to anchor its representations to absolute token indices via positional embeddings, spatial structure can be re-established through the fixed positional signal. As a result, *SSDC recovery under RPI* serves as a probe for index-based spatial organization: models that rely on absolute positional information exhibit non-trivial SSDC despite permutation, whereas models that rely on content-based cues collapse to near-zero SSDC.

We evaluate this behavior across models trained with learned absolute positional embeddings (APE), sinusoidal encodings, rotary embeddings (RoPE), no positional embeddings (ablated), and under Random Permutation Training (RPT). RPT preserves the presence of positional embeddings while



(a) **Gaussian Blur ($\sigma = 2.5$).** Fragility scores under a mild perturbation that removes high-frequency detail while largely preserving global spatial structure. Differences between models are present but compressed, reflecting the weaker disruption of content-based cues.

(b) **JPEG Compression (quality = 5).** Fragility scores under a strong content-disrupting transformation. Models with positional encodings exhibit substantially lower fragility than ablated and RPT models, with RoPE achieving the lowest fragility overall.

Figure 3: **Robustness to distributional shifts.** Fragility scores across model variants under two perturbation regimes. The gap between models with and without a stable positional reference frame is most pronounced under strong content disruption (JPEG), while remaining consistent but attenuated under milder perturbations (Gaussian blur).

disrupting the consistency between token indices and spatial locations during training; we include it here for completeness, and analyze its implications for robustness in the following subsection.

Results: Models trained without positional embeddings exhibit a complete collapse of SSDC under RPI across all layers, confirming that their spatial structure is entirely content-based. Despite exhibiting non-trivial SSDC in the unpermuted setting (Section 5.2), this structure does not survive disruption of patch content, indicating that it is not anchored to token indices.

In contrast, all models trained with positional embeddings show substantial SSDC recovery under RPI, demonstrating the emergence of index-based spatial organization. However, the nature of this recovery differs across encoding schemes. For APE and sinusoidal embeddings, SSDC exhibits a rapid increase in early layers following permutation, reaching a peak within the first few layers before stabilizing or slightly decreasing. This behavior suggests that spatial structure is injected early in the network via additive positional signals.

RoPE models display a qualitatively different trajectory: SSDC increases more gradually and continues to grow with depth, without a pronounced early-layer peak. This indicates that positional information is integrated progressively throughout the network, consistent with its multiplicative incorporation into attention mechanisms.

RPT models, despite having positional embeddings present, fail to exhibit meaningful SSDC recovery under RPI, behaving similarly to fully ablated models. This demonstrates that the mere presence of positional embeddings is insufficient; a consistent mapping between token indices and spatial locations during training is required for index-based spatial organization to emerge.

Taken together, these results establish that positional embeddings induce a shift from content-based to index-based spatial organization, and that this shift depends critically on the stability of the positional reference frame rather than on the architectural presence of positional signals alone.

5.4 Robustness to Content-Preserving and Content-Disrupting Perturbations

Experimental Setup: To evaluate how spatial organization strategy influences robustness, we measure model performance under controlled distribution shifts that perturb image content while preserving global structure. We consider two transformations with distinct characteristics:

JPEG Compression: We apply aggressive JPEG compression (quality = 5), which introduces blocking artifacts and significantly degrades local texture statistics. This transformation strongly

disrupts content-based cues while preserving coarse spatial layout, making it a targeted probe of reliance on local visual features.

Gaussian Blur: We apply Gaussian blur with standard deviation $\sigma = 2.5$, which attenuates high-frequency information while largely preserving low-frequency spatial structure. Compared to JPEG compression, this constitutes a milder perturbation that degrades fine-grained detail without substantially altering global organization.

For each model, we compute the *Fragility Score* (FS), defined as the relative drop in accuracy under each transformation.

In addition to standard positional encoding (PE) variants, we include **Random Permutation Training (RPT)** as a critical control condition. As mentioned in the subsection before, RPT preserves the architectural presence of PEs but models trained under RPT cannot form a stable index-based spatial organization despite having access to positional signals. This allows us to isolate whether robustness mainly arises from the mere presence of PEs or from the emergence of a consistent positional reference frame.

Results: Under JPEG compression, we observe a clear separation between models that develop index-based spatial organization and those that do not. Models with positional encodings exhibit substantially lower fragility (APE and sinusoidal: $\sim$0.43, RoPE: $\sim$0.30) compared to ablated and RPT models ($\sim$0.66). This large gap indicates that the presence of a stable positional reference frame is the dominant factor governing robustness to severe content degradation. Furthermore, within PE-based models, RoPE consistently achieves lower fragility than APE and sinusoidal encodings, suggesting that the specific encoding mechanism can modulate robustness, albeit to a lesser extent than the presence of positional structure itself.

Under Gaussian blur, the same ordering of models is preserved but with attenuated differences (RoPE: $\sim$0.15, APE: $\sim$0.17–0.20, sinusoidal: $\sim$0.22, RPT: $\sim$0.25, ablated: $\sim$0.30). This compression of the fragility gap is consistent with the milder nature of the perturbation: because Gaussian blur preserves much of the global spatial structure, it provides a weaker test of reliance on content-based spatial cues. Consequently, while index-based models remain more robust, the advantage is less pronounced than under JPEG compression.

Taken together, these results support a two-level interpretation. First, the emergence of a stable positional reference frame (enabled by consistent positional embeddings and absent in both ablated and RPT models) is the primary determinant of robustness to distributional shifts. Second, the specific form of positional encoding introduces secondary variation within this regime, with RoPE exhibiting consistently lower fragility, potentially due to its ability to maintain spatial structure more effectively across depth.

Crucially, the poor robustness of RPT models demonstrates that the mere presence of positional embeddings is insufficient: robustness depends on the model learning a consistent mapping between token indices and spatial locations. This directly links the spatial organization strategy identified in earlier sections to downstream robustness behavior.

5.5 Linking Index-Based Spatial Organization to Robustness via Positional Scaling

Experimental Setup: To establish a causal link between spatial organization and robustness, we require a controlled intervention that selectively disrupts index-based spatial structure while preserving the rest of the model. We achieve this by scaling the magnitude of learned absolute positional embeddings (APE) at inference time by a factor $\alpha \in [0, 1]$, without retraining.

While this intervention operates on positional embeddings, our goal is not to study positional signal strength per se, but to use it as a mechanism to continuously degrade the model's *index-based spatial organization*. To measure the integrity of this organization, we evaluate Spatial Similarity Distance Correlation (SSDC) under Random Permutation at Inference (RPI), as introduced in Section 5.3. Under RPI, any recovered spatial structure must be anchored to token indices rather than content. We therefore interpret SSDC recovery as a proxy for the presence of index-based spatial organization.

Importantly, SSDC recovery approaching zero does not imply that the model is actively constructing a content-based spatial structure; rather, it indicates that index-based spatial organization is no longer preserved under permutation.

To summarize this behavior compactly, we define:

$$\Delta\text{SSDC} = \text{SSDC}_{layer1} - \text{SSDC}_{layer0},$$

which captures the immediate recovery of spatial structure after the first encoder block under RPI. Since SSDC at layer 0 is identically zero under permutation, ΔSSDC serves as a direct measure of early-layer index-based spatial organization.

We jointly analyze ΔSSDC and the Fragility Score (FS) across varying α. For clarity, we report representative magnitudes illustrating distinct regimes, with the full results provided in the Appendix.

Results: We observe a clear correspondence between the degradation of index-based spatial organization and the loss of robustness.

At high magnitudes ($\alpha \geq 0.9$), models exhibit strong SSDC recovery ($\Delta\text{SSDC} \approx 0.37$–$0.47$), indicating intact index-based spatial organization. In this regime, fragility remains relatively low and stable (FS ≈ 0.44–0.46), suggesting that robustness is preserved when spatial structure is intact.

As α decreases into an intermediate regime ($0.8 \geq \alpha \geq 0.5$), SSDC recovery drops sharply ($\Delta\text{SSDC} \approx 0.30 \rightarrow 0.02$), reflecting the progressive breakdown of index-based spatial organization. This degradation is accompanied by a pronounced increase in fragility (FS $\approx 0.48 \rightarrow 0.63$). Notably, the most significant increases in fragility occur precisely where SSDC recovery is actively decreasing, indicating that robustness degradation is tightly coupled to the loss of spatial structure.

Below a critical threshold ($\alpha \leq 0.4$), SSDC recovery collapses to zero ($\Delta\text{SSDC} \approx 0$), indicating that index-based spatial organization is no longer recoverable under permutation. In this regime, fragility continues to increase, but only marginally (FS $\approx 0.65 \rightarrow 0.685$). This suggests that once spatial organization is fully disrupted, further degradation in robustness is no longer driven by changes in spatial structure, but instead reflects secondary effects such as reduced representational quality or distribution mismatch induced by scaling.

A complementary effect is observed at high magnitudes: when index-based spatial organization is already fully intact, small reductions in α have limited impact on fragility. Together, these observations reveal three regimes: (1) a stable regime with intact spatial organization and low fragility, (2) a transition regime where spatial structure degrades and fragility increases sharply, and (3) a collapsed regime where spatial organization is absent and fragility plateaus.

Overall, these results provide causal evidence that robustness is governed by the presence of index-based spatial organization. Positional scaling serves only as a means of intervention; the observed changes in robustness track the degradation of spatial structure rather than the magnitude of the positional signal itself.

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